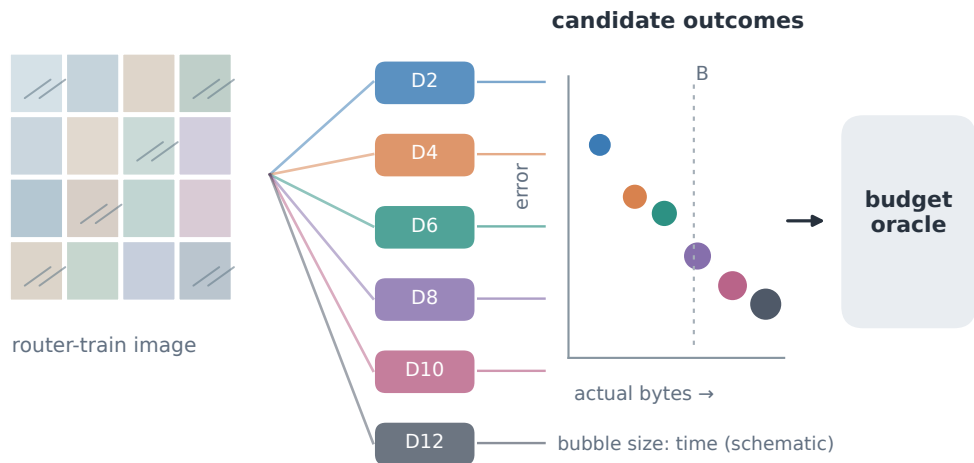


Learning decisions, then paying for only one path

Candidate outcomes are measured offline; deployment sees only the source and budget.

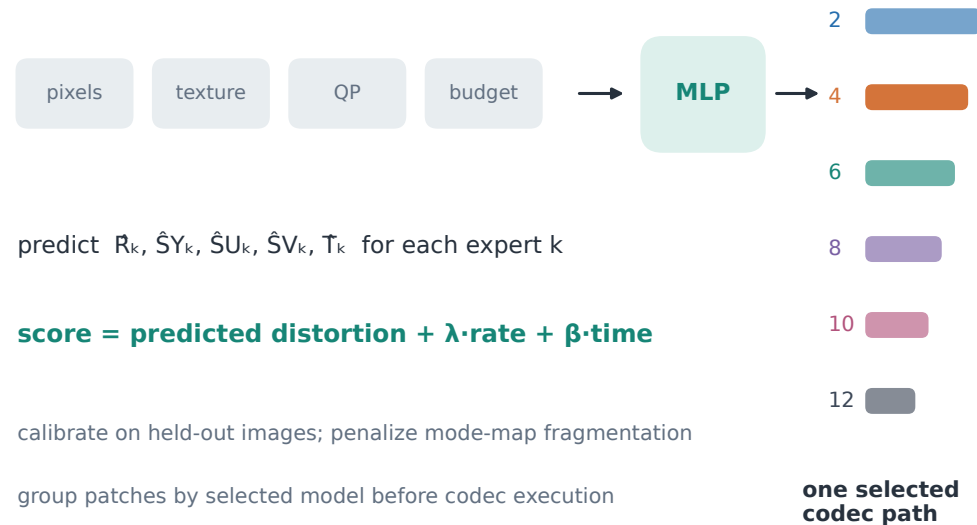
01 OFFLINE TARGETS



Measure actual bytes, Y/U/V error and time for all six paths.

Labels use the assembled image; split train/calibration/test by source image.

02 DEPLOYED DECISION



predict $R_k, \hat{S}Y_k, \hat{S}U_k, \hat{S}V_k, T_k$ for each expert k

$$\text{score} = \text{predicted distortion} + \lambda \cdot \text{rate} + \beta \cdot \text{time}$$

calibrate on held-out images; penalize mode-map fragmentation

group patches by selected model before codec execution

ACCOUNTING: router + selected encoder + entropy bytes + model IDs + selected decoder + tile assembly + transfers

Algorithm schematic; bar lengths and colors are illustrative, not measured quality or runtime. Full six-codec search is an offline upper bound and must be charged if run at inference.